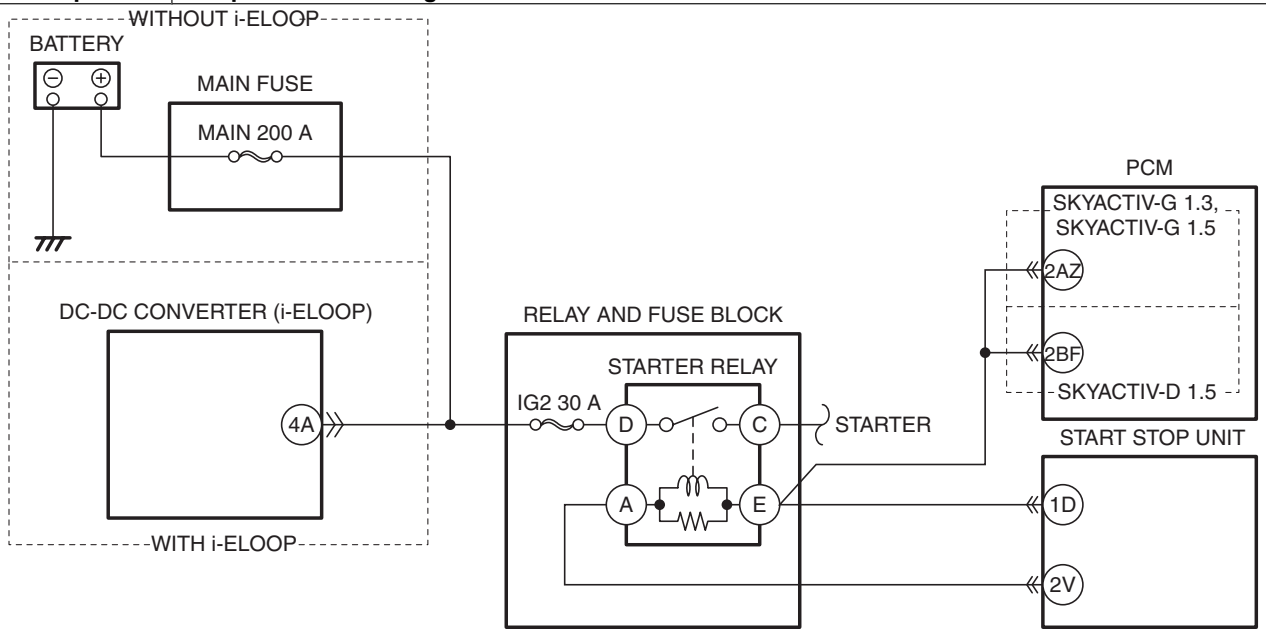
DTC B1140:29 [START STOP UNIT]

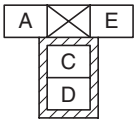
id0902p6020600

Description	i-stop control error signal received
Detection condition	• Start stop unit detects that the starter ground circuit does not turn on or open circuit on even though the PCM detects an engine restart request while the engine is stopped under i-stop control for **2 s or more** for a total of **5 times or more**.
Fail-safe	Not applicable
Possible cause	• DTCs are stored in the DC-DC converter (i-ELOOP). — DC-DC converter (i-ELOOP) connector or terminal malfunction — DC-DC converter (i-ELOOP) malfunction • Starter relay malfunction • PCM connector or terminal malfunction • Start stop unit connector or terminal malfunction • Open circuit in wiring harness between the following terminals: — SKYACTIV-G 1.3, SKYACTIV-G 1.5 • Starter relay terminal E—PCM terminal 2AZ • Start stop unit terminal 1D—Starter relay terminal E • Starter relay terminal A—Start stop unit terminal 2V — SKYACTIV-D 1.5 • Starter relay terminal E—PCM terminal 2BF • Start stop unit terminal 1D—Starter relay terminal E • Starter relay terminal A—Start stop unit terminal 2V • Start stop unit malfunction

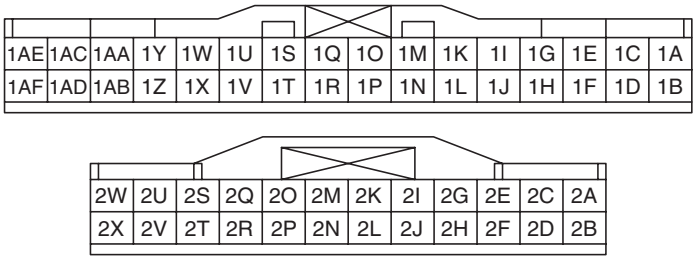
Description	i-stop control error signal received

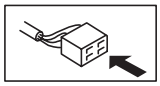


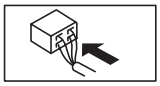
STARTER RELAY
(RELAY AND FUSE BLOCK)



START STOP UNIT
WIRING HARNESS-SIDE CONNECTOR

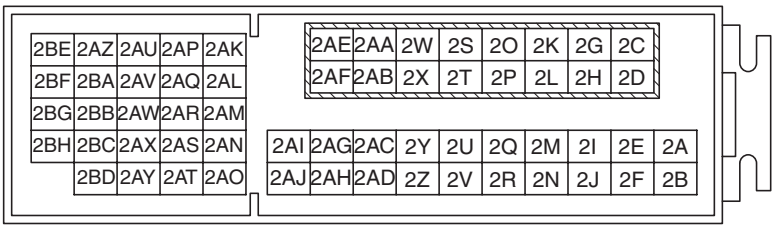




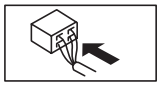


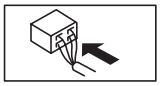
DC-DC CONVERTER (i-ELOOP)
WIRING HARNESS-SIDE CONNECTOR

PCM
WIRING HARNESS-SIDE CONNECTOR









Diagnostic Procedure

Step	Inspection		Action
1	**VERIFY DC-DC CONVERTER (i-ELOOP) DTCs** **Note** • For vehicles without i-ELOOP, go to the next step. • Retrieve the DC-DC converter (i-ELOOP) DTCs using the M-MDS. (See DTC INSPECTION [i-ELOOP].) • Are any DTCs displayed?	Yes	Repair or replace the malfunctioning part according to the applicable DTC troubleshooting. (See DTC TABLE [i-ELOOP].)
		No	Go to the next step.
2	**INSPECT STARTER RELAY** • Switch the ignition off. • Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Remove the starter relay. (See RELAY LOCATION.) • Inspect the starter relay. (See RELAY INSPECTION.) • Is the starter relay normal?	Yes	Go to the next step.
		No	Replace the starter relay, then go to Step 6. (See RELAY LOCATION.)
3	**INSPECT PCM CONNECTOR CONDITION** • Disconnect the PCM connector. • Inspect the connector engagement and connection condition and inspect the terminals for damage, deformation, corrosion, or disconnection. • Is the connector normal?	Yes	Go to the next step.
		No	Repair or replace the connector, then go to Step 6.
4	**INSPECT START STOP UNIT CONNECTOR CONDITION** • Disconnect the start stop unit connector. • Inspect the connector engagement and connection condition and inspect the terminals for damage, deformation, corrosion, or disconnection. • Is the connector normal?	Yes	Go to the next step.
		No	Repair or replace the connector, then go to Step 6.
5	**INSPECT STARTER RELAY CIRCUIT FOR OPEN CIRCUIT** • Verify that the starter relay is removed. • Verify that the PCM and start stop unit connectors are disconnected. • Inspect for continuity between the following terminals (wiring harness-side): — SKYACTIV-G 1.3, SKYACTIV-G 1.5 • Starter relay terminal E—PCM terminal 2AZ • Start stop unit terminal 1D—Starter relay terminal E • Starter relay terminal A—Start stop unit terminal 2V — SKYACTIV-D 1.5 • Starter relay terminal E—PCM terminal 2BF • Start stop unit terminal 1D—Starter relay terminal E • Starter relay terminal A—Start stop unit terminal 2V • Is there continuity?	Yes	Go to the next step.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between the following terminals: • SKYACTIV-G 1.3, SKYACTIV-G 1.5 — Starter relay terminal E—PCM terminal 2AZ — Start stop unit terminal 1D—Starter relay terminal E — Starter relay terminal A—Start stop unit terminal 2V • SKYACTIV-D 1.5 — Starter relay terminal E—PCM terminal 2BF — Start stop unit terminal 1D—Starter relay terminal E — Starter relay terminal A—Start stop unit terminal 2V **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. Go to the next step.

Step	Inspection		Action
6	**VERIFY THAT REPAIRS HAVE BEEN COMPLETED** • Always reconnect all disconnected connectors. • Connect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Clear the DTC for the start stop unit using the M-MDS. (See CLEARING DTC [START STOP UNIT].) • Perform the work of stopping the engine by i-stop control and then restarting the engine for **5 times or more**. • Retrieve the start stop unit DTCs using the M-MDS. (See DTC INSPECTION [START STOP UNIT].) • Is the same DTC displayed?	Yes	Repeat the inspection from Step 1. • If the malfunction recurs, replace the start stop unit. (See START STOP UNIT REMOVAL/INSTALLATION.) Go to the next step.
		No	Go to the next step.
7	**VERIFY IF OTHER DTCs DISPLAYED** • Are any other DTCs displayed?	Yes	Repair or replace the malfunctioning part according to the applicable DTC troubleshooting. (See DTC TABLE [START STOP UNIT].)
		No	DTC troubleshooting completed.